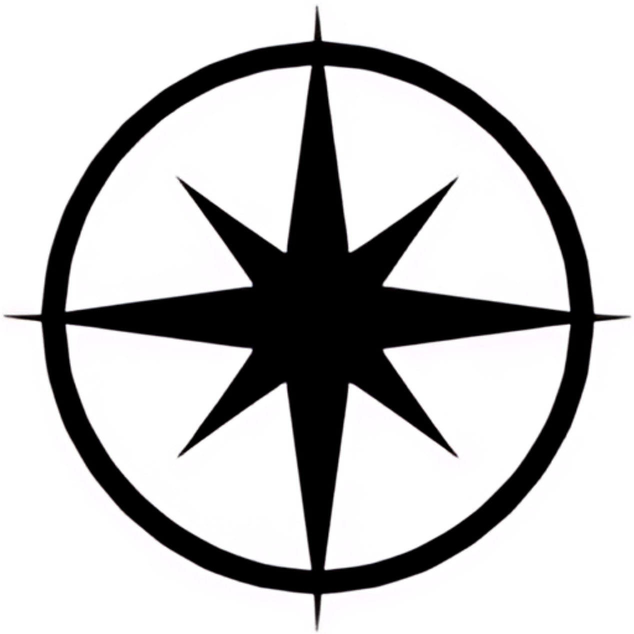

Tier 1:

1) $\frac{n(n+1)}{2}$

$(n+1) = \frac{(n+1)(n+2)}{2}$

$\frac{n(n+1)}{2} + (n+1) = \frac{n(n+1)+2(n+1)}{2}$

$= \frac{(n+2)(n+1)}{2}$ ✓

2) $n(n+1)$

$(k+1)(k+2)$

$k(k+1) + 2(k+1)$

$(k+2)(k+1)$ ✓

3) $n(n+1)$

$2(k+1)-1 = 2k+2-1 = 2k+1$

$(k+1)(k+2)$

$k(k+1) + 2k+1 = k(k+1) + 2(k+1)$

$(k+2)(k+1)$

Tier 2

4) $$\frac{n(n+1)(n+2)}{3} = \frac{(k+1)(k+2)(k+3)}{3}$$

$(k+1)(k+2)$

$$\frac{k(k+1)(k+2)}{3} + (k+1)(k+2)$$

$$\frac{k(k+1)(k+2)}{3} + \frac{3(k+1)(k+2)}{3}$$

$$\frac{K(K+1)(K+2)+3(K+1)(K+2)}{3}$$

$$\frac{(K+3)(K+1)(K+2)}{3} \checkmark$$

5) 2^n-1 $\quad 2^{K+1}-1$

2^{K+1-1}

2^K-1+2^K

$2^{K+1}-1 \checkmark$

6) $\frac{n}{n+1}$ $\quad \frac{K+1}{K+2}$

$$\frac{1}{(K+1)(K+2)}$$

$$\frac{K}{K+1}+\frac{1}{(K+1)(K+2)}$$

$$\frac{k(k+2)}{(k+1)(k+2)} + \frac{1}{(k+1)(k+2)}$$

$$\frac{k(k+2)+1}{(k+1)(k+2)} = \frac{k^2+2k+1}{(k+1)(k+2)}$$

$$\frac{(k+1)^2}{(k+2)(k+1)} = \frac{k+1}{k+2}$$

Tier 3

7) $$\frac{n^3+2n}{3}$$

$$k^3+2k = 3q$$

$$(k+1)^3+2(k+1) = 3q$$

$$(k+1)\ k^2+2k+1 = k^3+2k^2+k+k^2+2k+1$$

$$k^3+3k^2+3k+1 \quad +2k+2$$

$$k^3+3k^2+5k+3 = k^3+3k^2+2k+3k+3$$

$$(k^3+2k) + 3k^2+3k+3$$

$$3q+3k^2+3k+3$$

$3(q+k^2+k+1)$

let $x=q+k^2+k+1$

$3x$ ✓

8) $n^3-n=6q$ $\quad k^3-k=6q$

$(k+1)^3-(k+1)$

$(k+1)(k^2+2k+1)$

$k^3+2k^2+k+k^2+2k+1$

$k^3+3k^2+3k+1-k-1$

k^3+3k^2+2k

k^3+3k^2+3k-k

$(k^3-k)+3k^2+3k$

$6q+3k^2+3k$

$6(q+\frac{1}{2}k^2+\frac{1}{2}k)$ ✓

9) $7^n - 2^n = 5q$

$7^{k+1} - 2^{k+1}$

$7 \cdot 7^k - 2^k \cdot 2 = (7^k - 2^k)(7-2)$

$5q(7-2)$

$5q(5) \quad 25q$

$5(5q)$ ✓

10) $3^{2n} - 1 = 8q$

$3^{2k} - 1$

$9^k - 1 = 8q$

$3^{2(k+1)} - 1$

$9^{(k+1)} - 1 = 9 \cdot 9^k - 1$

$9^k + 8 \cdot 9^k - 1$

$8q + 8 \cdot 9^k$

$8(q + 9^k)$

Tier 4

11)
12) WTH there all hard
13)

Tier 5

14) $n^2+11n-12$ $\quad (K+1)^2+11(K+1)-12$

$K^2+2K+1+11K+11-12$

K^2+13K

$T_K+2(K+1)+10$

$K^2+11K-12+2K+2+10$

K^2+13K ✓

15) $7 \cdot 4^{k+1} - 2^{k+1}$

$7 \cdot 4^{k} - 2^{k}$

$4T_k + 2^{k+1}$

$4(7 \cdot 4^{k} - 2^{k}) + 2^{k+1}$

$7 \cdot 4^{k+1} - 4 \cdot 2^{k} + 2^{k+1}$

$7 \cdot 4^{k+1} - 2 \cdot 2^{k}$

$7 \cdot 4^{k+1} - 2^{k+1}$ ✓

16) original $= \frac{1}{4}(3^{k+1} - 2k^2 + 8k + 17)$

$k+1 = \frac{1}{4}(3^{k+2} - 2(k+1)^2 + 8(k+1) + 17)$

$= 3^{k+2} - 2(k^2 + 2k + 1) + 8k + 25$

$\frac{1}{4}(3^{k+2} - 2k^2 - 4k - 2 + 8k + 25)$

$= \frac{1}{4}(3^{k+2} - 2k^2 + 4k + 23)$

$3T_k + (k+1)^2 - 7(k+1) - 1$

$$\frac{3}{4}\left(3^{k+1} - 2k^2 + 8k + 17\right) + (k+1)^2 - 7(k+1) - 1$$

ugh 16 is hard